

Question 1: What is a variable in C/C++?

A variable is a named storage location in memory used to store data that can be changed during program execution.

Question 2: Why do we use variables in programs?

We use variables to store user inputs, keep track of data, and reuse values throughout our program.

Question 3: What is the difference between a variable name and its value?

Variable Name: It is the label used to identify the memory location (e.g., age).

Variable Value: It is the actual data stored inside that location (e.g., 20).

Question 4: Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks

b) total_marks is the only valid variable name.

Question 5: Why is Age different from age?

Because C++ is a case-sensitive language, so it treats uppercase and lowercase letters differently.

Question 6: Can a variable name contain spaces? Why?

No, a variable name cannot contain spaces because the compiler uses spaces to separate words and commands, which will cause a syntax error.

Question 7: Can a variable name start with an underscore (_) ?

Yes, a variable name can start with an underscore.

Question 8: What happens if two variables have the same name in the same scope?

The compiler will give a redefinition error (compile-time error) because it won't be able to identify which variable to use.

Question 9: Write three meaningful variable names for storing student information.

student_age

roll_number

total_marks

Question 10: Which naming convention is better and why: studentMarks or sm?

studentMarks is better because it is meaningful and easy to understand for anyone reading the code.

Question 11: What is a data type?

A data type tells the compiler what type of data a variable can store and how much memory it will take.

Question 12: Which data type is used to store whole numbers?

int data type.

Question 13: Which data type is used to store decimal values?

float or double data type.

Question 14: What is the difference between float and double?

float takes 4 bytes of memory and has less precision.

double takes 8 bytes of memory and has double the precision of a float.

Question 15: Which data type stores a single character?

char data type.

Question 16: Which data type stores True/False values?

bool data type.

Question 17: What data type is suitable for storing a person's name?

string data type.

Question 18: Predict the output: `int age=18; cout<<age;`

Output: 18

Question 19: Identify the data type: 25, 3.14, 'A', true.

25 is int

3.14 is double

'A' is char

true is bool

Question 20: What happens if you store a decimal number in an int variable?

The numbers after the decimal point are removed (truncated). For example, 5.7 becomes 5.

Question 21: What is the purpose of cin?

It is used to take input from the user (via keyboard).

Question 22: What is the purpose of cout?

It is used to display output on the screen.

Question 23: Which symbol is used with cin?

Extraction operator (>>).

Question 24: Which symbol is used with cout?

Insertion operator (<<).

Question 25: Explain: cin >> age;

It takes an input value from the user and stores it in the variable age.

Question 26: Explain: cout << age;

It prints the value stored in the variable age on the screen.

Question 27: What is the purpose of endl?

It is used to insert a new line and move the cursor to the next line.

Question 28: Write a statement to input a student's marks.

C++

```
cin >> marks;
```

Question 29: Write a statement to display 'Welcome to C++'.

C++

```
cout << "Welcome to C++";
```

Question 30: Predict the output: `cout << 'Programming';`

Output: It will give an error because single quotes are only used for a single character, not for whole words.

Question 31: What is an operator?

An operator is a symbol that tells the compiler to perform a specific mathematical or logical task.

Question 32: Which operator is used for addition?

+ operator.

Question 33: What is the result of `10 % 3`?

Result: 1 (because 10 divided by 3 leaves a remainder of 1).

Question 34: What is the result of `8 % 2`?

Result: 0 (because 8 is completely divisible by 2).

Question 35: Which operator checks equality?

== operator.

Question 36: What is the difference between = and == ?

= is used to assign a value to a variable.

== is used to compare if two values are equal.

Question 37: What is the result of $5 > 3$?

Result: true (or 1).

Question 38: What does && represent?

Logical AND operator (returns true only if both conditions are true).

Question 39: What does || represent?

Logical OR operator (returns true if at least one condition is true).

Question 40: What does ! represent?

Logical NOT operator (reverses the result, makes true into false and vice versa).

Question 41: Write a program to print 'Hello World'.

C++

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    cout << "Hello World";
```

```
    return 0;
```

```
}
```

Question 42: Write a program to input two numbers and display their sum.

C++

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int a, b, sum;  
    cout << "Enter two numbers: ";  
    cin >> a >> b;  
    sum = a + b;  
    cout << "Sum is: " << sum;  
    return 0;  
}
```

Question 43: Write a program to calculate the area of a rectangle.

C++

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int length, width, area;  
    cout << "Enter length and width: ";  
    cin >> length >> width;  
    area = length * width;  
    cout << "Area is: " << area;  
    return 0;  
}
```

Question 44: What formula is used to calculate Simple Interest?

Simple Interest = (Principal * Rate * Time) / 100

Question 45: Write a program to calculate Simple Interest.

C++

```
#include <iostream>

using namespace std;

int main() {
    float p, r, t, si;
    cout << "Enter Principal, Rate and Time: ";
    cin >> p >> r >> t;
    si = (p * r * t) / 100;
    cout << "Simple Interest is: " << si;
    return 0;
}
```

Question 46: Why is a temporary variable used while swapping two numbers?

Because if we directly copy one variable to another, the old value will be overwritten and lost. A temporary variable holds the value safely during the swap.

Question 47: Swap the values: a=10, b=20.

C++

```
#include <iostream>

using namespace std;

int main() {
    int a = 10, b = 20, temp;
    temp = a;
    a = b;
    b = temp;
    cout << "After swap: a = " << a << ", b = " << b;
```

```
    return 0;  
}
```

Question 48: What formula is used to convert Celsius into Fahrenheit?

Fahrenheit = (Celsius * 9/5) + 32

Question 49: What is the Fahrenheit value of 0°C?

32°F

Question 50: Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Because doing 9 / 5 using integers gives 1 instead of 1.8 in C++. It cuts off the decimal part, which makes the whole formula calculation wrong.